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KECK SPECTRA OF PLEIADES BROWN DWARF CANDIDATES AND A PRECISE DETERMINATION OF THE LITHIUM DEPLETION EDGE IN THE PLEIADES[1]

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ABSTRACT

We have obtained intermediate resolution spectra of 11 candidate brown dwarf members of the Pleiades open cluster using the Keck II telescope and Low-Resolution Imaging Spectrograph. Our primary goal was to determine the location of the "lithium depletion edge" in the Pleiades and hence to derive a precise age for the cluster. All but one of our 11 program objects have radial velocities appropriate for Pleiades members, have moderately strong Hα emission, and have spectral types M6–M8.5 as expected from their $(R - I)_C$ colors. We have constructed a color-magnitude diagram for the faint end of the Pleiades main sequence, including only stars for which high signal-to-noise spectra in the region of the lithium 6708 Å absorption line have been obtained. These data allow us to accurately determine the Pleiades single-star lithium depletion edge at $I_{C0} = 17.80$, $(R - I)_{C0} = 2.20$, spectral type M6.5. By reference to theoretical evolutionary models, this converts fairly directly into an age for the Pleiades of $\tau = 125$ Myr. This is significantly older than the age that is normally quoted, but does agree with some other recent estimates.

Subject headings: stars: low mass, brown dwarfs — open clusters and associations: individual (Pleiades)

1. INTRODUCTION

The Pleiades was recognized in the 1980s as the best open cluster for an attempt to identify brown dwarfs (Stauffer et al. 1989; Jameson & Skillen 1989) because of its fortuitous combination of proximity, youth, and richness. A number of brown dwarf candidates were identified in those and subsequent papers, usually on the basis of photometry obtained from deep imaging surveys in two or more colors. A means to establish that a candidate brown dwarf was at least near, if not necessarily below, the hydrogen-burning mass limit was proposed by Rebolo, Martín, & Magazzù (1992, hereafter RMM). Below about 0.065 $M_\odot$, brown dwarfs should never develop core temperatures sufficient for lithium ignition, and thus for objects less massive than this we should find a lithium abundance the same as for the interstellar medium from which the object formed, independent of the object's age. For slightly higher masses, lithium acts as an age scale because the length of time it takes for the core to reach 3.0×10^6 K is a sensitive function of mass. Because these stars are fully convective, once the core temperature exceeds the necessary limit, the entire lithium content of the star should be exhausted rapidly and thus be reflected in an observable change in the photospheric lithium abundance. The current generation of theoretical models make specific predictions about the time evolution of this lithium depletion boundary. For example, F. D'Antona & I. Mazzitelli (1997, private communication) predict that at ages 30, 70, and 140 Myr, the lithium depletion edge should occur at 0.17, 0.09, and 0.07 $M_\odot$, respectively. Other recent models by I. Baraffe et al. (1998, private communication) and Burrows et al. (1997) make nearly identical predictions of the variation of this lithium depletion boundary with age. Indeed, Bildsten et al. (1997) and others have argued that the age for an open cluster derived in this manner should be more accurate than those found by any other method.

The initial attempts to detect lithium in very low mass open cluster members were not successful (RMM; Marcy, Basri, & Graham 1994, hereafter MBG94). However, Basri, Marcy & Graham (1996, hereafter BMG96) eventually detected lithium in PPL 15, a Pleiades brown dwarf candidate originally identified by Stauffer, Hamilton, & Probst (1994). Rebolo et al. (1996) later showed that two other Pleiades members about 1 mag fainter than PPL 15 also have strong lithium absorption. BMG96 derived an age for the Pleiades based on the presence of lithium in PPL 15 but the absence of lithium in another only slightly brighter Pleiades member (HHJ 3). However, Basri & Martín (1998) have recently provided evidence that suggests that PPL 15 is a nearly equal-mass spectroscopic binary and thus its individual components would be ~0.75 mag fainter. This allows the location of the lithium boundary to be considerably fainter than had been assumed. The published data therefore no longer constrain the age of the Pleiades nearly as well as one would like because too few stars have been measured spectroscopically in the magnitude range of interest.

In this paper we report on the results of a program to determine lithium abundances for a number of candidate Pleiades members with apparent magnitudes chosen to bracket the possible magnitude range within which the lithium depletion boundary might be located.

2. OBSERVATIONS AND DATA REDUCTION

We obtained spectra of our Pleiades brown dwarf candidates with the Keck II Low-Resolution Imaging Spectrograph (LRIS) on the nights of 1997 December 3–5. We used a 1200 line mm^{-1} grating and a 1″ slit. The resultant dispersion was 0.63

[1] Based on observations obtained at the W. M. Keck Observatory, which is operated jointly by the University of California and the California Institute of Technology.

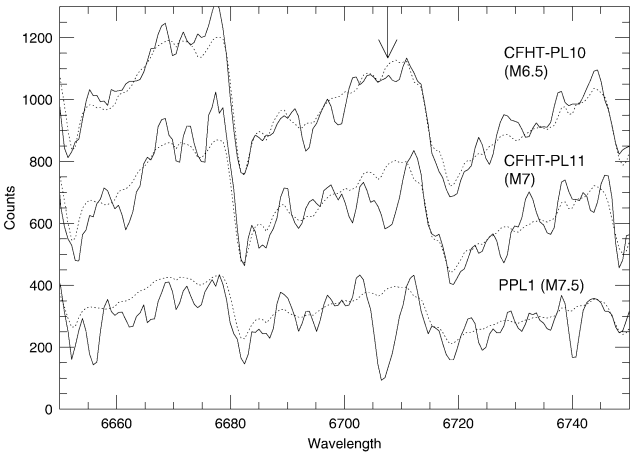


FIG. 1.—Sample spectra of Pleiades brown dwarf candidates obtained with the Keck II LRIS. The displayed wavelength region is only a small portion of the full spectrum, selected in order to highlight the lithium 6708 Å region. The *y*-axis is correct for CFHT PL 10, while the spectra of the other two stars are offset relative to CFHT PL 10 to avoid having the spectra overlap. The dashed line is a spectrum of GL 65AB, a field M6–M6.5 binary, assumed to have entirely depleted its initial lithium.

Å pixel^{-1}, and the measured spectral resolution was about 2.0 Å. The wavelength region observed was 6375–7650 Å.

The primary source from which we selected target objects was the recently completed deep imaging survey of the Pleiades conducted by Bouvier et al. (1998). Bouvier et al. identified 18 objects from that survey as good candidate Pleiades brown dwarfs based on their location in an I_C versus $(R - I)_C$ color-magnitude diagram. We obtained Keck spectra of seven of those candidates, with *I* magnitudes chosen to bracket the expected location of the lithium depletion edge. In addition, we obtained spectra of one target each from four other Pleiades brown dwarf imaging surveys (Stauffer et al. 1989; Schultz 1998; Stauffer et al. 1998a; Zapatero-Osorio et al. 1997), again selecting objects whose *I* magnitudes indicated they would help define the lithium depletion edge.

To first order, all of our spectra look the same: they have strong TiO and VO molecular bands indicative of spectral types M6–M8, and all of the stars we consider to be likely Pleiades members have moderately strong Hα emission. They do show differences in their lithium abundances, however. Figure 1 shows a closeup of the lithium 6708 Å region for three increasingly cooler Pleiades members, illustrating that at M6.5 we do not detect lithium at good signal to noise, that lithium is detected as a weak feature at M7, and that at later spectral types it is generally seen quite strongly in absorption.

We have used our spectra to provide the following quantitative information for our target objects: (1) Hα and lithium equivalent widths; (2) spectral indices that measure the continuum shape from 7000 to 7500 Å (PC2; see Martín, Rebolo, & Zapatero-Osorio 1996) and the depth of the VO band centered around 7450 Å (Kirkpatrick, Henry, & Simons 1995); and (3) radial velocities derived both from Gaussian fits to the Hα emission feature (the strongest, best-defined feature in the spectra) and from cross-correlation of the region from 6580 to 7050 Å versus GL 905, a dM5.5 standard.

Table 1 provides the relevant photometric and spectroscopic data for our target stars. Columns (2) and (3) provide I_{C0} and $(R - I)_{C0}$ photometry from Bouvier et al. (1998) for the Canada-France-Hawaii Telescope (CFHT) stars and from the papers cited above for the other stars, where the subscript zero indicates that we have corrected the observed photometry for an assumed reddening of $A_I = 0.06$ and $E(R - I) = 0.03$ (Stauffer & Hartmann 1987). Columns (4) and (5) are the PC2 and VO spectral indices, while column (6) is our spectral type derived from these indices using the stars in Kirkpatrick et al. (1995) and Martín et al. (1996) as standards. We have used Gliese catalog M dwarfs to define a linear relation between PC2 and $(R - I)_C$: $(R - I)_C = 0.746 \times \mathrm{PC2} + 0.65$ (see Stauffer et al. 1998b for details). These spectroscopic $R - I$ colors are provided in column (7) of Table 1. The last two columns of the table are the measured lithium and Hα equivalent widths. We will report the radial velocities and some other information derived from these spectra, including lithium abundances, in a subsequent paper (Stauffer et al. 1998b).

In the next section, we will present a color-magnitude diagram for the late-type Pleiades members for which there exist high-quality lithium equivalent width estimates. We cannot do this using measured photometric colors because the appropriate photometry does not exist [that is, some of the stars have $(R - I)_C$ colors, some have $I - K$ colors, but there is no single color where all of the stars have measured photometry]. However, all of the stars have $(R - I)_C$ colors estimated from spectroscopy or have spectral indices from which $(R - I)_C$ colors can be inferred. We therefore will use spectroscopic $(R - I)_C$ colors in that figure. Comparison of the photometric and spectroscopic estimates of $(R - I)_C$ for the seven CFHT stars in Table 1 suggests that this is an acceptable procedure.

It is also necessary to know whether the objects we have observed are Pleiades members or not. As discussed in Bouvier et al. (1998), the probability that a photometric candidate from

TABLE 1
DATA FOR PLEIADES BROWN DWARF CANDIDATES

Name (1)	I_{C0} (2)	$(R - I)_{C0}$ (3)	PC2 (4)	VO (5)	Spectral Type (6)	$(R - I)_{calc}$ (7)	EW(Li) (Å) (8)	EW(Hα) (Å) (9)
CFHT PL 9	17.65	2.15	2.03	1.05	M6.5	2.17	<0.05	3.8
CFHT PL 10	17.76	2.18	2.03	1.04	M6.5	2.17	<0.05	5.4
CFHT PL 11	17.85	2.18	2.13	1.06	M7	2.24	0.5	5.1
CFHT PL 12	17.94	2.44	2.51	1.11	M8	2.52	0.8	17.2
CFHT PL 13	17.96	2.18	2.16	1.06	M7	2.26	0.6	8.7
CFHT PL 14	17.98	2.14	1.87	1.03	M6	2.05	<0.1	0
CFHT PL 15	18.56	2.31	2.14	1.07	M7	2.25	0.5	7.0
Roq 13	18.19	...	2.31	1.08	M7.5	2.38	2.5	4.8
MHObd3	18.21	...	2.46	1.11	M8	2.49	0.8::	5.9
PPL 1	17.65	...	2.42	1.11	M7.5	2.46	2.4	6.5
Lick PL 1	17.14	...	2.05	1.03	M6.5	2.18	<0.1	5.3

the CFHT survey will indeed be a Pleiades member, as estimated simply from the expected field star contamination given the area and depth surveyed, is fairly good: ~70% or better. If in addition the object has a spectral type that agrees with the photometric color, has Hα emission strength that matches what is expected for very low mass Pleiades members (Stauffer, Liebert, & Giampapa 1995), and has detected lithium—indicating an age less than 1 Gyr for stars in this spectral type range—then we believe that the object is almost certainly a Pleiades member. Nondetection of lithium fainter than the I_C magnitude where other Pleiades members have detected lithium would suggest that the object is a nonmember. For brighter candidate Pleiades members, lithium is not useful but radial velocities can provide further evidence in support of Pleiades membership. In fact, all of the stars in Table 1 except CFHT PL 14 have radial velocities compatible with Pleiades membership within the expected accuracy of our measurements (~4 km s^{-1}, 1 σ, for the velocities derived from Hα, ~8 km s^{-1}, 1 σ, for the cross-correlation velocities). CFHT PL 14 also lacks Hα emission and has no measurable lithium feature. We therefore assume that all of the stars in Table 1 are Pleiades members except for CFHT PL 14, which we assume is a nonmember.

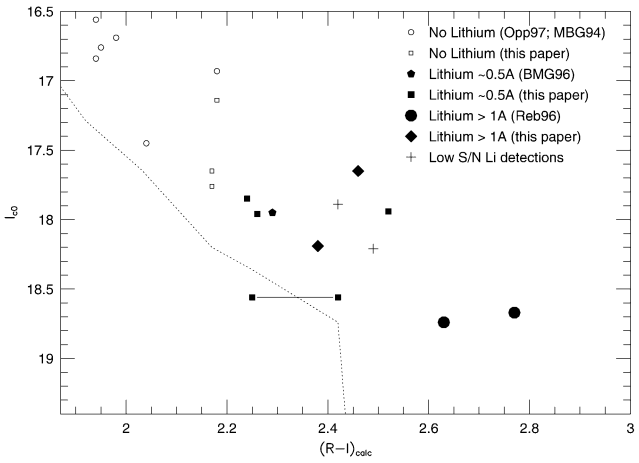


FIG. 2.—Color-magnitude diagram showing only those very low mass Pleiades members for which a good assessment of their lithium abundance can be made. The dashed line is an empirical ZAMS shifted to Pleiades distance. The *x*-axis colors are inferred from our spectroscopy (and so are reddening free); the *y*-axis has been corrected for an assumed reddening of $A_I = 0.06$. The two "low signal-to-noise detections" are MHObd3 (see Table 1) and MHObd1 (see Stauffer et al. 1998a).

3. DISCUSSION

The goal of this project was to precisely determine the location of the lithium depletion edge in the Pleiades, and hence to determine an accurate age for the cluster. Figure 2 shows a color-magnitude diagram for very low mass Pleiades members for which lithium data are available. The lithium data are from Oppenheimer et al. (1997, hereafter Opp97), MBG94, BMG96, Rebolo et al. (1996, hereafter Reb96), and this paper. The $(R - I)_C$ colors for the Opp97 stars are from Stauffer et al. (1995), where we have converted these spectroscopic $(V - I)_C$ colors to $(R - I)_C$ via a relation derived from the Gliese catalog M dwarfs in Leggett (1992). The $(R - I)_C$ colors for some stars in Reb96 and BMG96 are from PC2 indices provided by Martín et al. (1996) and our calibration of PC2 versus $(R - I)_C$. Two points for CFHT PL 15 are shown: $(R - I)_C = 2.24$ as derived from its PC2 index, and $(R - I)_C = 2.41$ as derived from its VO index. Using either color, CFHT PL 15 lies below the main sequence defined by the other stars, possibly indicating that it is a nonmember. Because it has detected lithium and a radial velocity compatible with Pleiades membership, we prefer to believe it is a member and that the inferred color is anomalous (either intrinsically or because of measurement error). Our primary conclusions are unaffected by how we interpret this star. Finally, the dashed line in Figure 2 is a field star ZAMS derived from the M dwarf photometry provided by Leggett (1992).

The location of the stars in Figure 2 show a well-defined correlation with the measured lithium equivalent widths. Bluer than $(R - I)_C = 2.2$, none of the stars have detected lithium, while redder than that color, all of the measured stars have lithium. Similarly, fainter than $I_{C0} = 17.8$, all the stars have detected lithium, whereas brighter than that limit, all but one of the stars are without lithium. Finally, there is a trend for lithium equivalent width to increase going to lower inferred mass (i.e., fainter and redder). We further believe that the dispersion in *I* magnitude at a given color is to a large extent real and is primarily an indication that some of the observed stars are photometric binaries. In particular, we suggest that HHJ 6 ($I_{C0} = 16.93$, $R - I_C = 2.18$), Lick PL 1, PPL 1, and CFHT PL 12 are good candidates to be photometric binaries. CFHT PL 12 also has considerably stronger Hα emission than the other Pleiades stars observed, possibly indicating it is a short-period binary or that we have observed it during a flare. The attribution of PPL 1 as a nearly equal-mass binary would explain why it has strong lithium absorption despite an I_C magnitude equal to or brighter than two other cluster members with no detected lithium. An alternative explanation would be that there is a significant age spread in the Pleiades and these four overluminous stars are the youngest in our sample. By comparison to theoretical models (e.g., F. D'Antona & I. Mazzitelli 1997, private communication), their displacement about 0.5 mag above other stars of the same color would require them to be more than 50 Myr younger than the other stars, which we believe is unlikely (see, for example, discussions of this issue by Soderblom et al. 1993 and Steele & Jameson 1995).

Based on the above interpretation of Figure 2, we determine that the single-star lithium depletion edge in the Pleiades is at $I_{C0} = 17.8 \pm 0.1$ or $(R - I)_{C0} = 2.20 \pm 0.05$. The uncertainty estimates are not rigorous, and arise mostly from the uncertainties in the absolute calibration of the photometry in Bouvier et al. (1998). For theoretical evolutionary models which incorporate realistic model atmospheres as their outer boundary condition, and hence which can predict observational colors and magnitudes for brown dwarfs, it is possible to convert directly the empirical lithium depletion edge to an age estimate for the Pleiades. In Figure 3 we plot the absolute I_C magnitude of the lithium depletion boundary (defined here as the point where lithium has been depleted by a factor of 100) for the most recent models of I. Baraffe et al. (1998, private communication) as a function of age. To place our empirically measured point into this diagram, we assume that the cluster has $(m - M)_0 = 5.60$ ($r \sim 130$ pc), and $A_I = 0.06$ (see Pinsonneault et al. 1998), leading to $M(I_C) = 12.2 \pm 0.15$ for the lithium depletion boundary, where we have assumed plausible but again unrigorous 1 σ uncertainties of 0.1 mag for the distance modulus and 0.03 mag for the extinction and that the uncertainties add in quadrature. The age derived in this way is then 125 ± 8 Myr, where the uncertainty estimate only comes from propagating the 0.15 mag uncertainty of the boundary through the model shown in Figure 3.

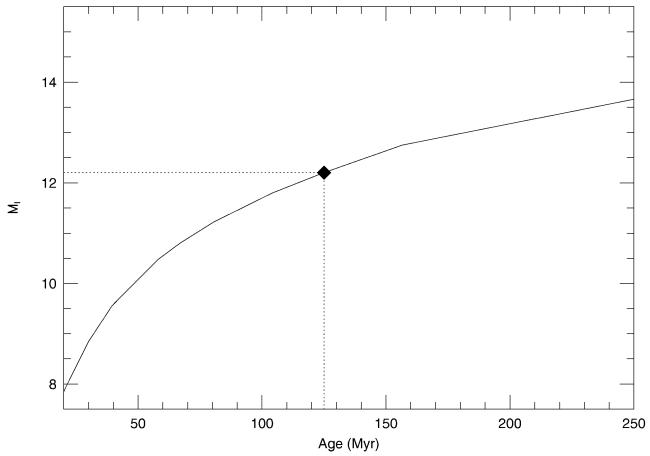


FIG. 3.—Predicted absolute I_C magnitude where stars should have depleted 99% of their initial lithium as a function of age based on the models of I. Baraffe et al. (1998, private communication). The diamond symbol marks our empirical location of this lithium depletion boundary in the Pleiades, indicating an age for the cluster of 125 Myr.

In order to try to assess the model dependence of this estimate, we have also made a similar calculation for theoretical evolutionary models by Burrows et al. (1997) and F. D'Antona & I. Mazzitelli (1997, private communication). Those models do not provide R and I magnitudes, so instead we have used the $(R - I)_C$ color to estimate an I_C-band bolometric correction [we adopted the Monet et al. 1992 BC_I vs. $(V - I)_C$ relation, and a conversion from $(V - I)_C$ to $(R - I)_C$ based on data in Leggett 1992). In that manner, we estimate that the lithium depletion boundary in the Pleiades is at $M(\mathrm{bol}) = 11.99$. Comparing this number to the predictions of the two theoretical models, we get an age for the Pleiades of 130 Myr based on the calculations of F. D'Antona & I. Mazzitelli (1997, private communication) and 125 Myr for the Burrows et al. (1997) models.

The most commonly quoted age for the Pleiades is of order 70–80 Myr (Patenaude 1978; Mermilliod 1981). However, models with a relatively large amount of convective core overshoot can yield much larger ages, as was originally shown by the models of the Padova group (see Mazzei & Pigatto 1989, who derived an age for the Pleiades of 150 Myr). Other recent models give ages intermediate between these values (e.g., 100 Myr for Meynet, Mermilliod, & Maeder 1993 and ≥120 Myr for Ventura et al. 1998). Given the disagreement over the age derived from the upper–main-sequence turnoff, it is particularly useful to have an independent means to derive the age from low-mass stars. Using the lithium detection in PPL 15 and the nondetection of lithium in HHJ 3, BMG96 estimated the age of the Pleiades to be 115 Myr. Our new result pushes this age even slightly older; more important, it does so using many more stars and thus provides a much better defined age from the lithium data.

Finally, we note that we have chosen to use the "traditional" distance scale to the Pleiades ($r \sim 130$ pc). That distance conflicts with the new *Hipparcos* distance to the Pleiades of about 116 pc (van Leeuwen & Ruiz 1997; Mermilliod et al. 1997). We have done this because we believe that the *Hipparcos* distance for the Pleiades is not correct, as has been discussed in Pinsonneault et al. (1998) and Soderblom et al. (1998). The zeroth-order effect of simply using the *Hipparcos* distance and keeping everything else the same would lead to an even older Pleiades age of about 140 Myr.

4. SUMMARY AND IMPLICATIONS

Using the Keck II telescope, we have obtained good intermediate-resolution spectra of 10 very low mass probable members of the Pleiades. We detect lithium in seven of these stars, bringing the total number of confirmed lithium brown dwarfs in the Pleiades to 10. By also having spectra of several slightly brighter Pleiades members which do not show lithium, we have accurately determined the lithium depletion edge in the Pleiades to be at $M(I_C) = 12.2$. Using three different theoretical evolutionary models, we derive a "lithium age" for the Pleiades of 125–130 Myr. The age of the Pleiades as derived from the upper–main-sequence turnoff is variously quoted as between ~75 and ~150 Myr. If our new age as derived from location of the lithium depletion boundary is correct, and if the Pleiades is normal in some sense, our result suggests that the ages for many young open clusters should be shifted toward the old end of the scale, in approximate accord with the age scale derived from the models of the Padova group (Mazzei & Pigatto 1988) or that of Ventura et al. (1998).

Because the Pleiades stars have been extensively studied, they have served as one of the cornerstones in defining how rotational velocities, lithium abundances, coronal activity, etc., evolve with time. For most of those studies, an age of 75–100 Myr has been adopted for the Pleiades. Shifting the age of the Pleiades to 125 Myr (with or without shifting the ages of the other well-studied nearby clusters) will require modifications to our view of how these various properties evolve with time. This in turn will then affect the theoretical models that have been developed to explain these observed trends.

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